

PHY-7012 – APPLIED ASTRODYNAMIC & SSA

Course outline and instructor presentation

Richard L. Lachance, Ph.D.

SpaceOrbitLAB.com

RLLac@ULaval.ca



SpaceOrbitLAB



In collaboration with
ABB Space and
Defence Systems

Space Sciences

0. Introduction:

- Plan de cours, matériel, lien utiles
- Déroulement, calendrier, modules

1. Structure des modules

2. Généralités sur le cours

3. Présentation du professeur

4. Programmation scientifique

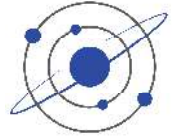
- MATLAB/Octave → Python

5. Références & outils de recherche

6. Acronymes



PHY-7012 A2026 – Projets Spéciaux = Astrodynamique SSA



www.ulaval.ca/etudes/cours/phy-7012-sujets-speciaux-physique

Université Laval > Études > Cours >

Version: 2025-05-09 09:00:00

PHY-7012

Sujets spéciaux (physique)

Cours thématique à sujet variable qui permet d'offrir, de manière ponctuelle et exceptionnelle, une formation particulière à un ou plusieurs étudiants.

Responsables

Faculté des sciences et de génie

Département de physique, de génie physique et d'optique

Restrictions à l'inscription

Cycle d'études

Doit être inscrit à:

- Deuxième cycle
- Troisième cycle

Certaines sections de cours peuvent comporter des restrictions additionnelles.

Des questions sur ce cours?

Inscrivez-vous aux cours

3
Crédits

Cycles du cours
Deuxième cycle
Troisième cycle

Modes
d'enseignement
Régulier

À l'horaire
Automne 2026

Horaire

Pour vous inscrire, accédez à [monPortail](#).

Automne 2026 – 1 section offerte

PHY-7012

H

Hybride

Restrictions

NRC 87825

Capacité maximale: 10 étudiants

Enseignant: Richard L. Lachance

Plages horaires

Type:	Sur Internet	Dates:	Du 31 août 2026 au 11 déc. 2026
Type:	En classe	Dates:	Du 31 août 2026 au 11 déc. 2026
Journée:	Lundi	Horaires:	09 h à 11 h50

Restrictions à l'inscription

Cycle d'études

Doit être inscrit à:

- Deuxième cycle
- Troisième cycle

Cette page constitue la description officielle de cette activité. L'Université Laval se réserve le droit de modifier l'activité sans préavis. Tous les horaires indiqués sont sujets à changement.



FIRST PAGE

MASTER THE PHYSICS OF MOTION: From first principles to real missions

An astrodynamics course for engineers,
analysts and researchers

Discover the fundamental principles governing
the motion of satellites, planets, and spacecraft
through a clear and progressive learning experience

[Explore courses →](#)

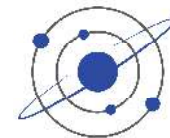
[Discover the Program](#)

[Free Python notebook](#)



In collaboration with
ABB Space and Defence
Systems

Autumn Calendar 2026



www.ulaval.ca/sites/default/files/notre-universite/direction-gouv/Documents_officiels/Calendrier_universitaire/Calendrier_universitaire_2025-2026.pdf

[Course schedule, pre-requisites, detailed outline & pricing – Advanced Astrodynamics & Space Situational Awareness \(SSA\) – Autumn 2026 2026-09-03.docx](#)

→ The course is now on Tuesday AM

August

S	M	T	W	T	F	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	31					

September

S	M	T	W	T	F	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30			

14 weeks

Pause on reading week : 25–31 October

October

S	M	T	W	T	F	S
						1 2 3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
Pause →	25	26	27	28	29	30 31

November

S	M	T	W	T	F	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30					

December

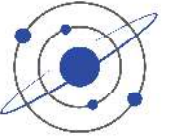
S	M	T	W	T	F	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

→ Recommended exercises after each lecture

→ 3 Homeworks

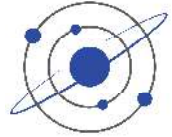
→ 1 Capstone projects

Schedule A2026



- Schedule: August 31 to December 8, 2026 (14 weeks, 26 classes, ≈30 hours of instruction)
 - [COPL-3156-Z, Tuesday 09:00 am to noon](#)
- Course Format
 - Blended: In-person and Teams teleconference (dates announced in advance)
 - Quizzes, Exercises, Assignments, and Exams/Capstone Projects
- All lectures are recorded as modules via Teams and available for replay
 - Theory & Practice (scientific coding)
- In-person classes canceled due to storms or other reasons will be held via Teams teleconference
 - If necessary, students may review the recorded version of the missed class
- Information for students is available on the course web site: www.SpaceOrbitLAB.com
 - Note: Except for the student's personal use for educational purposes, **no part of these materials, including the Orbits101 simulation software, may be reproduced, distributed, or transmitted. All rights reserved ©**
- Between lectures, the instructor will be available via email for technical questions and comments
RLLac@ULaval.ca

Planned activities – Part 1

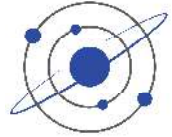


- web site link
(TODO)

#	Title	Date
i	Introduction – General concepts (Math & geometry)	2026-08-31
I	The Earth: geodesy and orientation into space + Time <i>review</i>	2026-09-07
II	Foundations of Astrodynamics <i>review</i> Astrodynamics laws, Orbital motion, Kepler problem, Reference frames & coordinate systems, Orbital elements, Ballistics	2026-09-14
III-1	Key Astrodynamics Concepts / Applied Astrodynamics 1. Positional astronomy – Ephemerides 2. Beta Angle and thermal considerations 3. Satellite in Earth Shadow 4. Orbital perturbations 5. Orbit propagators 6. Orbit Determination 7. Ballistic threat assessment	
	<i>Reading week</i>	2026-10-25

Detailed
schedule in the
making; will be
updated regularly

Planned activities – Part 2

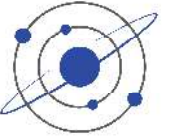


- web site link
(TODO)

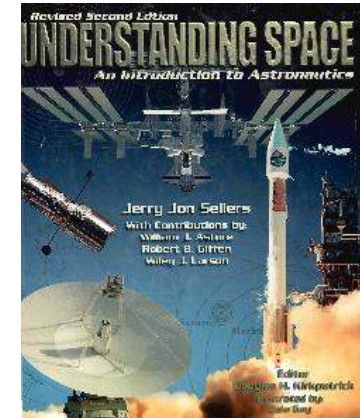
#	Title	Date
III-2	Space Situational Awareness & Space Domain Awareness	
	8.1 Space Debris 8.2 Debris Population Model 8.3 Orbital Environment Modeling 8.3 Hill / Clohessy-Wiltshire (HCW) equations 8.4 Covariance propagation 8.5 Collision probability + Conjunction assessment	
	Review & correction of capstone project	2026-12-07



Complementary educational materials



- Web Book, Rice University 2022 – *Astronomy 2e*
 - <https://openstax.org/details/books/astronomy-2e>
 - <https://openstax.org/books/astronomy-2e/pages/1-introduction>
 - [Download free pdf](#) openstax™
- Book “J.J. Sellers, *Understanding Space: An introduction to Astronautics*”
Revised 2nd Ed. → [PDF fourni](#)



“The most incomprehensible thing about the universe is that it is comprehensible.”

– *Albert Einstein, Physics and Reality, 1936.*

“Remember to look up at the stars and not down at your feet.

Try to make sense of what you see and wonder about what makes the universe exist. Be curious.”

– *Stephen Hawking*

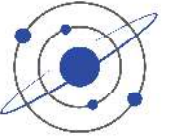
“Space has a magic way of inspiring people, and inspiring students in going into STEMs”

– *Pam Melroy, Deputy Administrator, NASA*

“Those who know nothing of the world of nature about them, or of the heavens above them, miss many of the intellectual and spiritual pleasures of life.”

– *Clarence Augustus Chant*

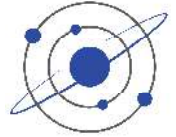
Goal of the course



*The present work is intended to inspire and excite the wonder in science students and tech professionals, to **fire their imaginations**, and to convey to them some notion of **the majesty, the mystery, and the sublimity of our Universe***



Space from Canada → Québec!



■ LinkedIn: [CSA Canadian Space Agency | Agence spatiale canadienne](#)

- [Student-built rocket](#)
- [Balloon launch](#)
- etc.

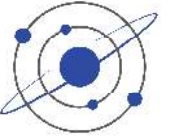


■ [GAUL](#) – Groupe Aérospatial de l'Université Laval



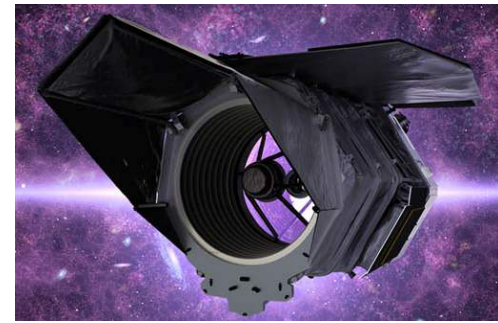
+2

Roman Space Telescope

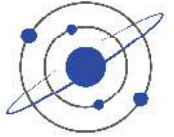


Canadian Space Agency | Agence spatiale canadienne

- The [Nancy Grace Roman Space Telescope](#) has been launched on August 30th, with Canadian technology on board and Canadian scientists contributing to its mission
- Two ultra-sensitive cameras developed by Canadian companies Nüvü Cameras and **ABB** are part of the mission. Capable of detecting very faint signals, they will help Roman search for distant worlds and advance science
- From cutting-edge technologies to research, Canadian expertise is helping Roman take a new look at the cosmos.
- <https://lnkd.in/gVpvJfGq> [ABB News \(fr\)](#) [ABB News \(en\)](#)



CSA SCISAT & ACE – Atmospheric Chemistry Experiment



Canadian Space Agency | Agence spatiale canadienne

109,619 followers

1w • Edited •

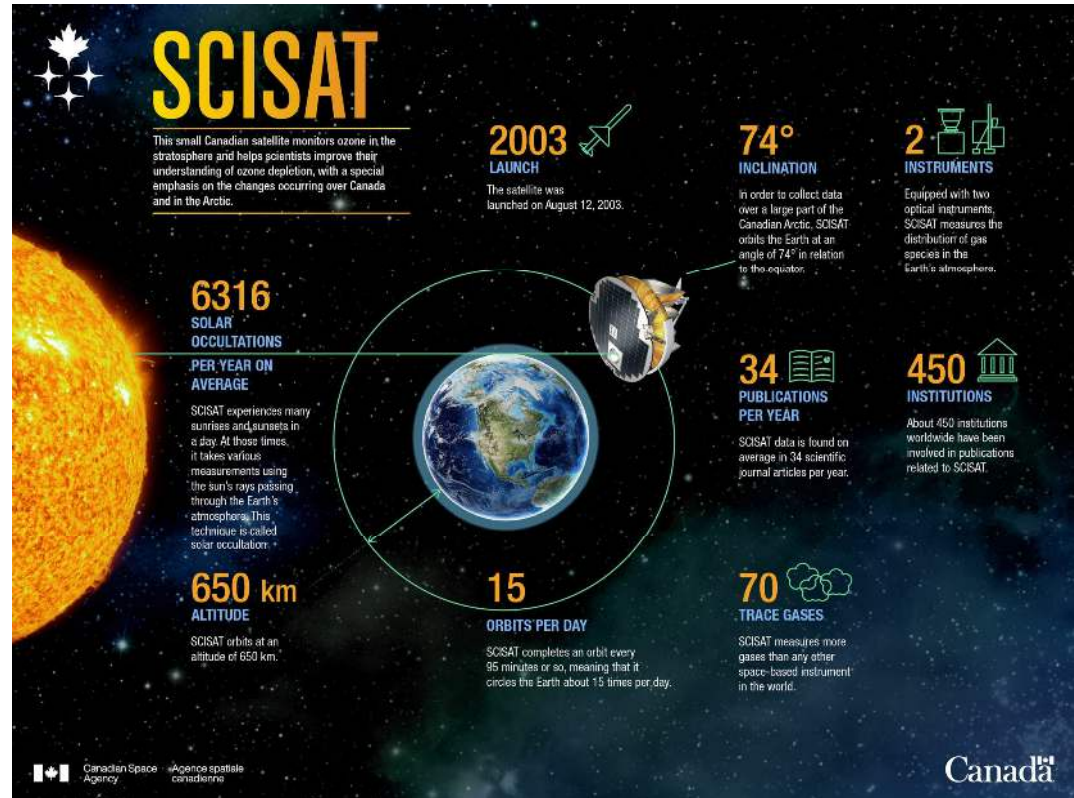
Today is International Day for the Preservation of the Ozone Layer – but for the Canadian satellite SCISAT, it's just another day on the job! 📡🇨🇦

- ☀️ Wakes up 650 km above Earth (no coffee needed). First scan? Ozone levels over the Arctic. Gotta check that protective layer!
- 🌍 Looks at Earth's atmosphere horizontally – the side view gives more detail on sneaky gases like CFCs, methane & more.
- 📡 Beams data to Earth: Canada, Europe, Japan. Sharing is caring (and climate-saving).
- 🌀 Swings over the tropics – time to measure how pollution moves around the globe and between layers of the atmosphere.
- 🌙 Still working. Because UV rays don't rest, and neither does SCISAT.

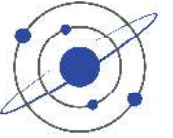
Launched in 2003, the satellite helps a team of national and international scientists monitor the recovery of the ozone layer, with a special emphasis on the changes occurring over Canada and in the Arctic.

www.asc-csa.gc.ca/eng/satellites/scisat/about.asp

#Satellite #PreservationOfTheOzoneLayer #Canada



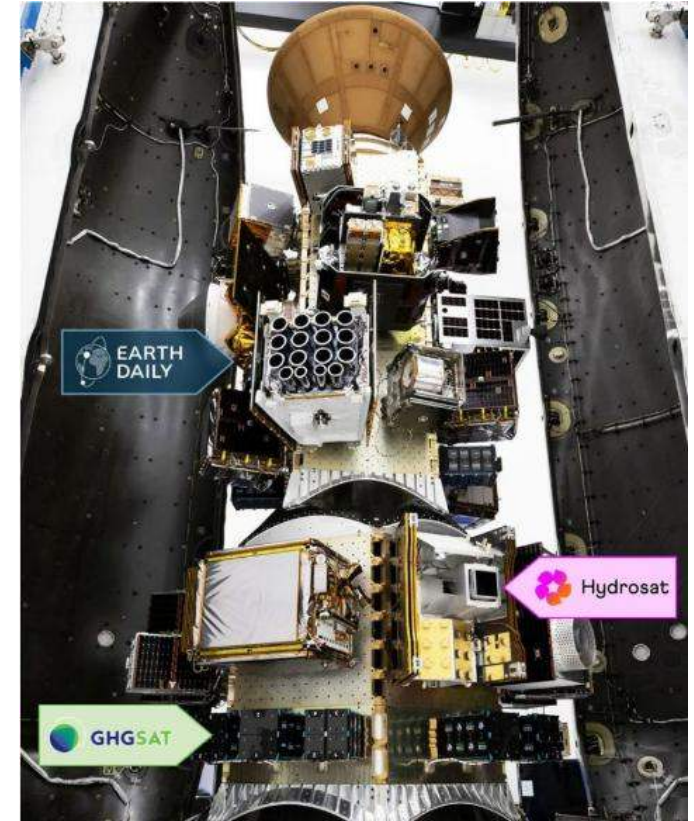
Mission example



- SpaceX Transporter-14 Rideshare Mission (2025-06-23 21:18 UTC) heading to Sun-synchronous orbit from Vandenberg California
- 45 customer satellites were deployed, carrying nearly 3000 kg of payload and launch hardware to LEO

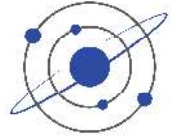
ABB: Very special day in our 30+ years space history!

- 4 satellites strapped on Transporter-14 are based on payloads built by ABB team in Quebec City:
 - Earthdaily FM-1 (marked as Loft Orbital's YAM-10)
 - GHGSat C-12 & C-13
 - Hydrosat VanZyl-2 (marked as Muon's MuSat-3)



In collaboration with
ABB Space and
Defence Systems

“A picture is worth a thousand words”



- This course covers different topics and lessons structured into a summarized theory, supported by practical examples and problems in the form of high-level algorithmic coding exercises

Physical phenomenon → **Descriptive model/equations** → **Program code/simulation** → **Figures** → **Understanding**

- Plotting an equation or a formula into a figure is worth a thousand explanations, and lets the student more deeply understand the physical phenomenon involved. This is why this course emphasize on visual demonstrations of each presented subject, with practical exercises simulating real cases.

Lesson subject → **Rotations: sign**

CODE: STUDY_Rotations.m

← separate code example

Explanation →

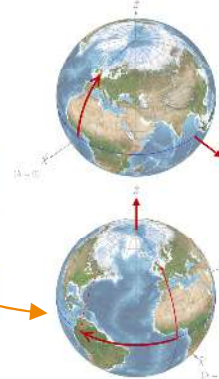
```
% ECEF (XYZ): Starting point at  $\lambda = 0^\circ$ ,  $\phi = 0^\circ$ 
Earth = AstroConstants();
xyz = Sph2Cart(0, 0, Earth.r1_km);
plot3(xyz(1,:), xyz(2,:), xyz(3,:), 'ro');

%--- Move up to latitude  $\phi = +51.5^\circ$  (Greenwich latitude)
Lat = +51.5; % [°N]
xyz_ = RotY(Lat) * xyz; % + requires a negative sign, as latitudes are
                        % measured clockwise, while the RotY rotation
                        % operator is defined counter-clockwise

plot3(xyz_(1,:), xyz_(2,:), xyz_(3,:), 'ro');

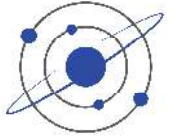
%--- Move westward to longitude  $\lambda = 71.2^\circ$ W (Quebec City longitude)
Lon = -71.2; % [°W] negative sign for westward longitudes
xyz_ = RotZ(Lon) * xyz; % + positive here, as longitude are measured
                        % counter-clockwise, the same as RotZ operator

plot3(xyz_(1,:), xyz_(2,:), xyz_(3,:), 'ro');
```



← validation figures,
allowing students to
explore further,
experimenting by
changing parameters,
rotating in 3D, etc.

Specialized course vs public astrodynamics libraries

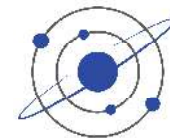


- Understanding the low-level implementation of astrodynamics libraries such as Astropy and Orekit is substantially more difficult than understanding astrodynamics itself from a textbook. The challenge comes not from the mathematics alone, but from the gap between the mathematical formulation and the software engineering practices used to build production-quality scientific software. Here are the main reasons:

1. The code is optimized for correctness, performance, and maintainability—not education

- Academic textbooks typically derive equations in a logical order:
 1. Physical assumptions
 2. Mathematical derivation
 3. Final equations
 4. Numerical examples
- Production software almost never follows this structure. Instead, the implementation is organized around:
 - minimizing memory allocations
 - avoiding unnecessary trigonometric evaluations
 - numerical stability, handling edge cases
 - compatibility with previous versions
 - reusable software architecture

Astropy & Orekit

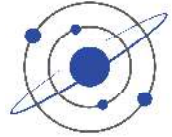


- [Astropy](#) and [Poliastro](#) are Python ecosystems designed primarily for **general observational astronomy and orbital mechanics & astrophysics**. In contrast, [Orekit](#) is a **low-level space dynamics library** written in Java, built specifically for highly accurate orbital mechanics, satellite operations, and flight dynamics

	Astropy	Orekit
Core Focus & Application	Focuses on analyzing observational data (images, spectra, light curves). It handles coordinate transformations between celestial reference frames, processes cosmological models, and manages physical units	Focuses on physical spacecraft. It models high-fidelity orbital maneuvers, atmospheric drag, solar radiation pressure, and Earth gravity fields
Typical Use Cases	Planning telescope observations, querying astronomical catalogs , and analyzing datasets from space telescopes	Space mission design , initial orbit determination, satellite conjunction assessment, and trajectory propagation
Platform and Interoperability	Native Python library that integrates easily with the standard Python data science stack (numpy, pandas, scipy)	Java library, though it is frequently utilized in Python environments using the Orekit Python Wrapper
Coordinate Systems & Time	Supports a wide variety of celestial coordinate frames geared toward viewing distant stars, galaxies, and the solar system	Tailored to Earth-centered orbits, supporting terrestrial and space-based reference frames and high-precision time scales (accounting for leap seconds and EOP)

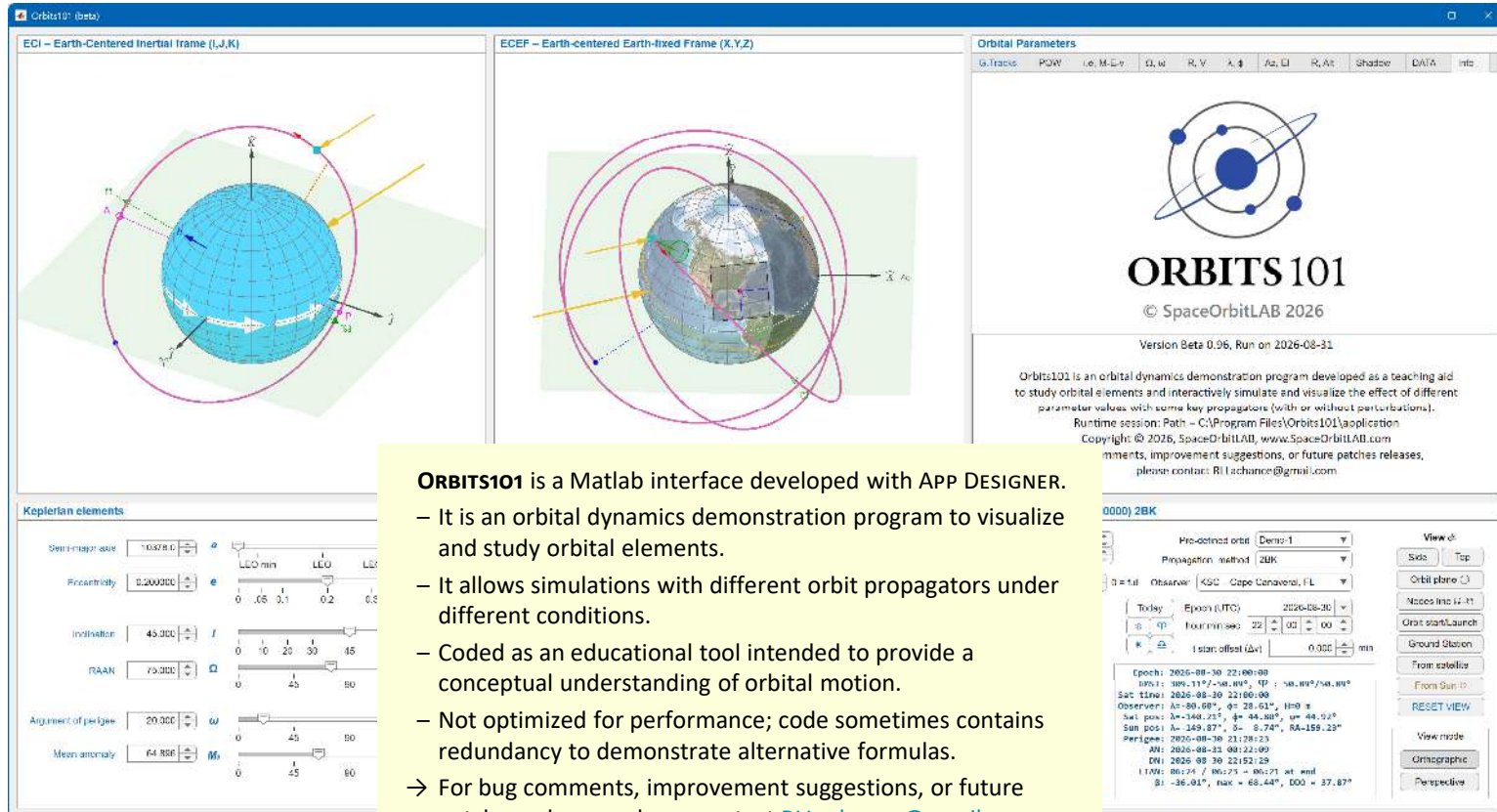
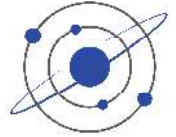
- While you can use libraries like Astropy to get the position of a planet at a specific time, you would use Orekit if you need to calculate the exact trajectory, fuel requirements, or atmospheric reentry of an artificial satellite.

Related software



- astropy:** “The Astropy Project is a community effort to develop a single core package for Astronomy in Python and foster interoperability between Python astronomy packages”. Not only it provides important core features for poliastro like time and physical units handling, but also sets a high bar for code quality and documentation standards.
- Skyfield:** Another Astronomy Python package focused on computing observations of planetary bodies and Earth satellites.
- Plyades:** A pioneering astrodynamics library written in Python. Its clean and user friendly API.
- orbital:** Yet another orbital mechanics Python library. It is very similar to poliastro but with smarter internal structure. It is more focused in plotting and it even provides 3D plots and animations.
- Orekit-python-wrapper:** “The Orekit python wrapper enables to use Orekit within a normal python environment”, using JCC. Orekit is a well-established, mature open source library for Astrodynamics written in Java strongly supported by several space agencies. The Python wrapper is developed by the Swedish Space Corporation
- beyond:** A young flight dynamics library written in Python with a focus on developing “a simple API for space observations”. Some parts overlap with poliastro, but it also introduces many interesting features, and the examples look promising
- SpiceyPy:** This [Python library wraps the SPICE Toolkit](#), a huge software collection developed by NASA which offers advanced astrodynamics functionality. Among all the wrappers available on the Internet, it is one of the most advanced and well-maintained one, although there are others

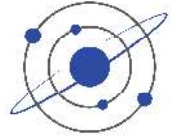
ORBITS101: Demonstration program used in the course



ORBITS101 is a Matlab interface developed with APP DESIGNER.

- It is an orbital dynamics demonstration program to visualize and study orbital elements.
 - It allows simulations with different orbit propagators under different conditions.
 - Coded as an educational tool intended to provide a conceptual understanding of orbital motion.
 - Not optimized for performance; code sometimes contains redundancy to demonstrate alternative formulas.
- For bug comments, improvement suggestions, or future patches releases, please contact RLlachance@gmail.com

Disclaimer

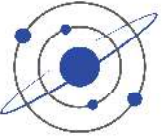


The content of this course is **intended solely for pedagogical purposes** and should not be applied directly to operational scenarios. The equations and physical constants used in this course **are for instructional purposes only** and may involve simplifications and/or approximations and should not be used as such for design, analysis or mission planning purposes.

While example screen shots resulting from analysis algorithms are presented, no endorsement for any code is made.

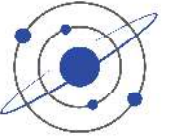
- The MATLAB and Python code provided is written with an emphasis on clarity rather than performance optimization, and as such, it is not guaranteed to be free of errors.
The code is a work in progress and subject to ongoing improvements.
- The author appreciates your understanding and encourage you to contact him if you identify any bugs or have suggestions for improvement. Your feedback is invaluable and helps enhance the quality of the material (e.g. removing typos and improving language in the present course).
- Additionally, if you'd like to stay informed about future document updates and software & code patch releases, please feel free to reach out.

Modifying the Source Code

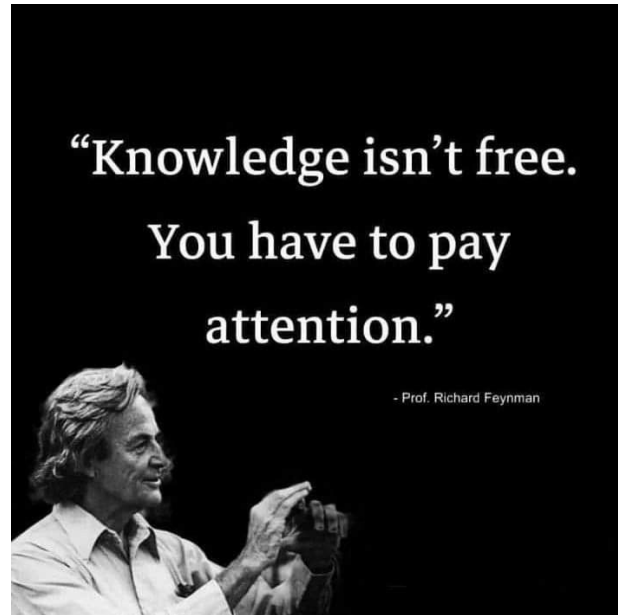


- ... the code is pedagogical, not optimized; some with alternative formulas and complementary algorithms
- ... Not for direct operational use
- The student/reader is free to modify the source code and to experiment with it.
- Indeed, there are even suggestions for such experimentation provided throughout the course. Under no circumstances, however, should the original or modified code be distributed. It is copyrighted and your purchase only entitles you to personal use without written permission from RLL Consulting
- Certain modifications have purposely not been made to the software so that the user can learn more by experimenting.

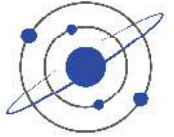
References and Credits



→ Unless otherwise credited, images, figures and animations in this course are all generated from MATLAB code developed by the author. Media attributions are given for other cases



(credit: modification of work by ...)

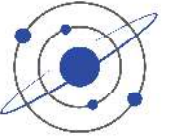


- Redistribution and use of text and in source code, with or without modification, are permitted, provided that the following conditions are met:
 - Redistributions of source code must retain the above copyright notice, this list of conditions and the following disclaimer.
 - Redistributions in binary form must reproduce the above copyright notice, this list of conditions and the following disclaimer in the documentation and/or other materials provided with the distribution
 - Neither the name of nor the names of its contributors may be used to endorse or promote products derived from this software without specific prior written permission.
- THIS SOFTWARE IS PROVIDED BY THE COPYRIGHT HOLDERS AND CONTRIBUTORS “AS IS” AND ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE ARE DISCLAIMED. IN NO EVENT SHALL THE COPYRIGHT OWNER OR CONTRIBUTORS BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

Instructor bio



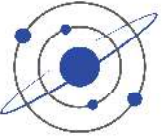
About the instructor



- Richard L. Lachance, Ph.D.

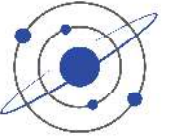


Richard L. Lachance high-level resume



- Ph.D. Optics (Université Laval 1994) & inventor of 8 patents
- 30+ years of working on [Earth remote sensing](#) technology for different spaces agencies (ESA, CSA, NOAA, NASA & JPL, JAXA) developing sensors in operation on [6 different satellite missions](#)
- Startup in telecom (optical components for fiber high speed internet) – 10 years, successful exit
- Remote sensing Business Development Specialist at ABB Space & Defence department
 - Specialist in atmospheric remote sensing business development, responsible for major government accounts and agencies like the NASA, NOAA, and EPA
 - Account Manager for [Department of Defence Canada](#) developing chemical warfare detectors (trace gas optical remote sensor) for protecting NATO troops
- Startup developing compact methane sensors (ground-based + drones) for the Oil & Gas industry ([Bluefield](#))
- Pivoted back to space with startup Bluefield in 2016, as co-founder and CTO
- Now acting as senior consulting for space missions in 2021, while teaching his career's experience

Richard's space related experience

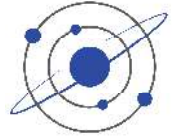


- ABB, 2011–2014, Business Development Specialist, Remote sensing
 - Specialist in [atmospheric remote sensing](#) business development for on-board aero-space and terrestrial technologies (satellites, balloons, and aircrafts).
 - Expand terrestrial environmental market and new applications related to [industrial leak detection](#), pollution measurements and other atmospheric applications having positive impact on the environment.
 - Responsible for atmospheric sales in US territory, especially for major government accounts and agencies like the NASA, NOAA, and EPA.
- Gas Plume Imaging (GPI) Canada Inc., 2010–2011, CEO
 - Developing advanced optical remote technologies to provide [trace gas monitoring services](#) for Oil & Natural Gas and related industries, using networks of ground-based and airborne sensors.
- Bomem, 1994–2000, Senior System Engineer, Scientist and Project Manager
 - [Remote sensing and space applications](#): modeling and analysis, optical instrumentation performance simulations, calibration and characterization. Instrument testing and validation. Ground segment algorithm design. Development of new technologies.

Startup Bluefield experience



Unlocking the Future of Natural Gas
Precise, always-on methane leak detection via microsatellites

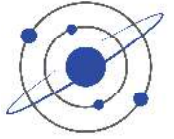


- Dr. Lachance served as the **Chief Technical Officer (CTO)** and **co-founder** of Bluefield Technologies, a climate-tech startup established in 2017 that developed advanced methane-leak detection sensors utilizing gas filter correlation radiometry (GFCR), a NASA-heritage optical technique.
- Under his technical leadership, Bluefield built compact airborne sensors capable of identifying methane emissions with high specificity, and conducted airborne demonstrations in 2018 in collaboration with oil and gas industry partners and environmental regulators. Lachance's tenure at Bluefield involved hands-on technical development, field validation efforts, and scaling a novel sensor platform, all within the challenging context of startup dynamics and capital-intensive environmental tech markets.
 - Despite the promise of these next-generation sensors, Bluefield ultimately ceased operations due to Covid-19 unforeseen challenges.



Sensor prototype for airborne demonstration; Flight tests over Québec City in December 2018

Startup advisor Chris Hadfield, Canadian astronaut



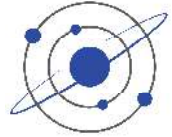
- Colonel [Chris A. Hadfield](#) has flown two Space Shuttle missions and also served as commander of the International Space Station. He acted as advisor for the trace gas detection startup Bluefield.



In session with Chris Hadfield, 2019



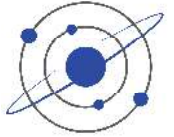
Experience with spacecraft development



Experience as Senior Scientist, Lead System Engineer and Program Manager in algorithms definition, system performance analysis & modeling, instrument simulation, test definition and ground calibration facility development

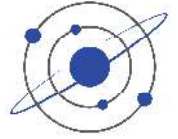
- **Envisat/MIPAS** – *Michelson Interferometer for Passive Atmospheric Sounding* – Operational in orbit 2002–2012
 - High-resolution Fourier transform spectrometer on the European Space Agency (ESA) ENVISAT satellite platform for the study of constituents of the atmosphere. One of the most massive and complex interferometer ever launched into space
 - Responsible for the operational algorithms; wrote the “Level 1B Algorithm Technical Baseline Document (ATBD)”
- **NPP/CrIS** – *Cross-track Infrared Sounder* – JPSS-1 Operational in orbit 2011–now, JPSS-2 in orbit 2017–now
 - NOAA Interferometer providing improved measurements of temperature and moisture in the Earth’s atmosphere. Forecasters use this data to help improve global & regional predictions of weather patterns and storm tracks.
 - System engineer for definition of calibration and data processing algorithm, system tests and performance analysis, user requirements, as well as specific ground support equipment.
- **Metop/IASI** – *Infrared Atmospheric Sounding Interferometer* – Metop-A operational in orbit 2006–now, B: 2012 –now
 - FTS for weather forecasting, the study of the ozone layer, the radiative interaction of clouds, and the study of oceans. Eumetsat/CNES instrument for the METOP series and the main instrument for supporting Numerical Weather prediction.
 - Project manager for the interferometer simulator: design and concept, characterization and test plan, performance verification, definition and participation in writing of the simulation software

Experience with spacecraft development



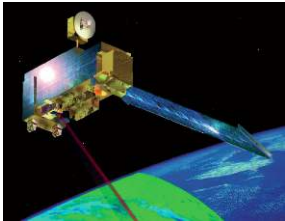
- **SciSat/ACE** – *Atmospheric Chemistry Experiment* – Operational in orbit 2003–present
 - Canadian Space Agency (CSA) scientific satellite providing spectral signatures of the atmosphere by solar occultation to study the chemical dynamics suspected of depleting the ozone in the stratosphere.
 - Originally designed for 2 years of operation, is still operating after 18 years due to robust design and engineering
 - Contributed to mission design planning, instrument concepts, and calibration algorithms
- **GOSAT-1/TANSO-FTS** – *Solar Occultation FTS for Inclined-orbit Satellite* – Operational in orbit 2009–present
 - The Greenhouse gases Observing SATellite “IBUKI” (GOSAT) is designed to measure the concentration of major greenhouse gases from space. This project is a joint effort promoted by the Japan Aerospace Exploration Agency (JAXA), the National Institute for Environmental Studies (NIES) and the Ministry of the Environment (MOE).
 - The TANSO-FTS instrument is a Michelson-type interferometer on-board GOSAT satellite features high optical throughput, fine spectral resolution, and a wide spectral coverage (from VIS to TIR in four bands).
- **Terra/MOPITT** – *Measurements of Pollution in the Troposphere* – Operational in orbit 1999–present
 - MOPITT is a nadir sounding (vertically downward pointing) instrument that measures upwelling infrared radiation at 4.7 μm and 2.2–2.4 μm . It uses correlation spectroscopy to calculate total column observations and profiles of carbon monoxide in the lower atmosphere. Funded by the Space Science Division of the Canadian Space Agency (CSA).
 - Worked on the calibration blackbodies

Richard's space missions experience



- Experience as Senior Scientist, Lead System Engineer and Program Manager
 - At mission-level, system & subsystem-levels

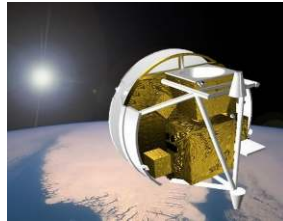
MOPITT – Terra
1998 – present



MIPAS – Envisat
2002 – 2012



ACE – SciSat-1
2003 – present



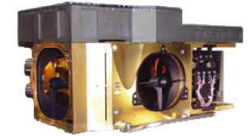
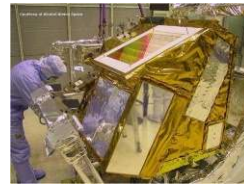
IASI – Metop
2006 – present



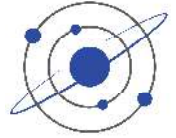
TANSO – Gosat-1
2009 – present



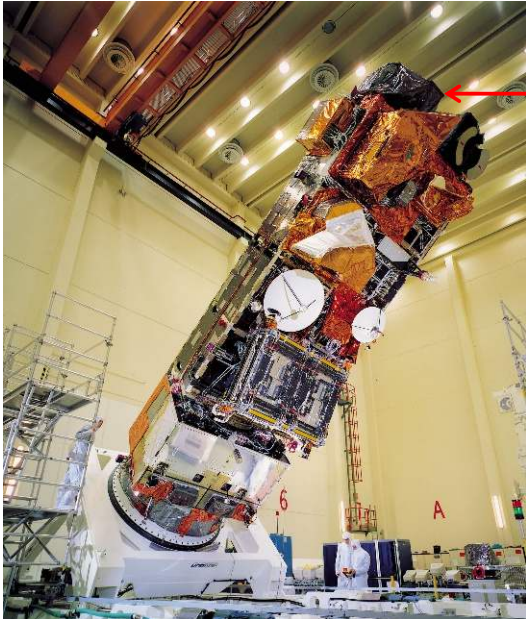
CrIS – Suomi-NPP
2011 – present



Collaboration Québec (BOMEM) – ESA/ESTEC



ENVISAT <https://earth.esa.int/eogateway/missions/envisat/description>



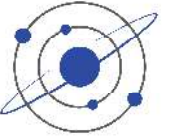
MIPAS: Michelson Interferometer for Passive Atmospheric Sounding

Integration of the complete Envisat/Polar Platform
Flight Model in ESA ESTEC center



RLL working on **MIPAS** interferometer alignment (ENVISAT)
DASA Deutsche Aerospace, Ottobrunn, München, 1996-10-28

To contact the instructor / author



Richard L. Lachance



www.linkedin.com/in/richard-l-lachance-28a6a4a/

SpceOrbitLAB.com (under construction)

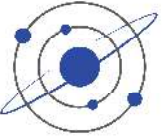
→ RLLAC@ulaval.ca / Richard-L.Lachance.1@ulaval.ca

Scientific programming environment

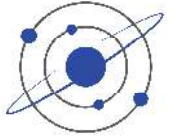
- MATLAB/Octave
- Python



Selection of a programming language



- Awareness of programming languages for astrodynamics computation helps decide when a different tool may be more appropriate for a specific task (MATLAB/Octave, Python, R, C++, Java, etc.)
- An important feature of **MATLAB/Octave** and **Python** is that they are an **interpreted language**.
 - This is advantageous because it enables **interactive programming** in a **Read–Eval–Print Loop (REPL)**: code entered into the console is immediately executed and the result is displayed, rather than waiting for the intermediate stage of compilation. IMP: **Both handle vectorial processing**
 - On the other hand, compiled languages such as C++ and Java tend to *run faster* (once they have been compiled).
- **Python** is a popular language for data science and is often used for astrodynamics. It is object-oriented and supports portable libraries to the benefit of the entire open source data science ecosystem.
- In practice both MATLAB/Octave and Python have their strengths.
 - To some extent which one is used is less important than the domain of application and communication of results. Learning either will provide a head-start in learning the other.
- C++ provides the basis for many libraries and GIS packages, and is better for final operational code
 - Well-written C++ is very fast, making it a good choice for performance-critical applications such as processing large geographic datasets, but is *harder to learn and master* than MATLAB/Octave or Python, especially at the **algorithm development level** compared to operational code programming.



Scientific-algorithmic programming

```

%% SOL_Finding_maximum
% My first example

x = linspace(0, 2*pi); % 100 pts by default
y = sin(x);

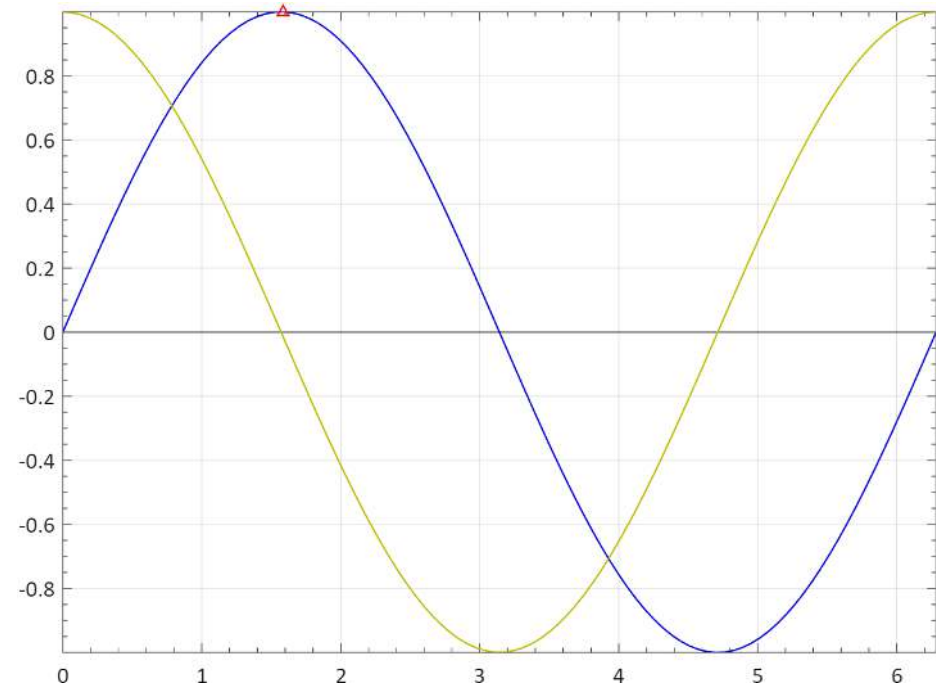
Figure(1, 'Sine wave');
h1 = plot(x, y);
Abscissa();

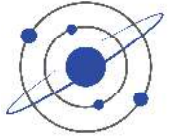
%--- Finding maximum

% 1) Numerical
[y_max, k1] = max(y); % k1 is vector index
% direct inspection
y_max == y(k1) % by definition
x(k1) % → 1.58666
plot(x(k1), y(k1), 'r^');

% 2) using derivative
dydx_ = Differentiate(x, y);
% here "_" denotes numerical solution
plot(x, dydx_);

```





Scientific-algorithmic programming

% Compares with theoretical function

dydx = cos(x); % $d \sin(x) / dx \equiv \cos(x)$

plot(x, dydx, 'm--');

% Finds zero in numerical derivative

[x1_cross, k2] = HitCross(x, dydx, 0); % (2 solutions)

[pi/2 , x1_cross(1) , x(k1)] 1.570796 1.570792 1.58666

plot(x1_cross, 0 * x1_cross, 'ro');

plot(x1_cross, y(k2), 'r+');

% Theoretical maximum position

plot(pi/2 * [1,1], ylim, 'k--');

if 0 % Zoomed region for study

xlim([pi/2 + 0.15 * [-1,1]]);

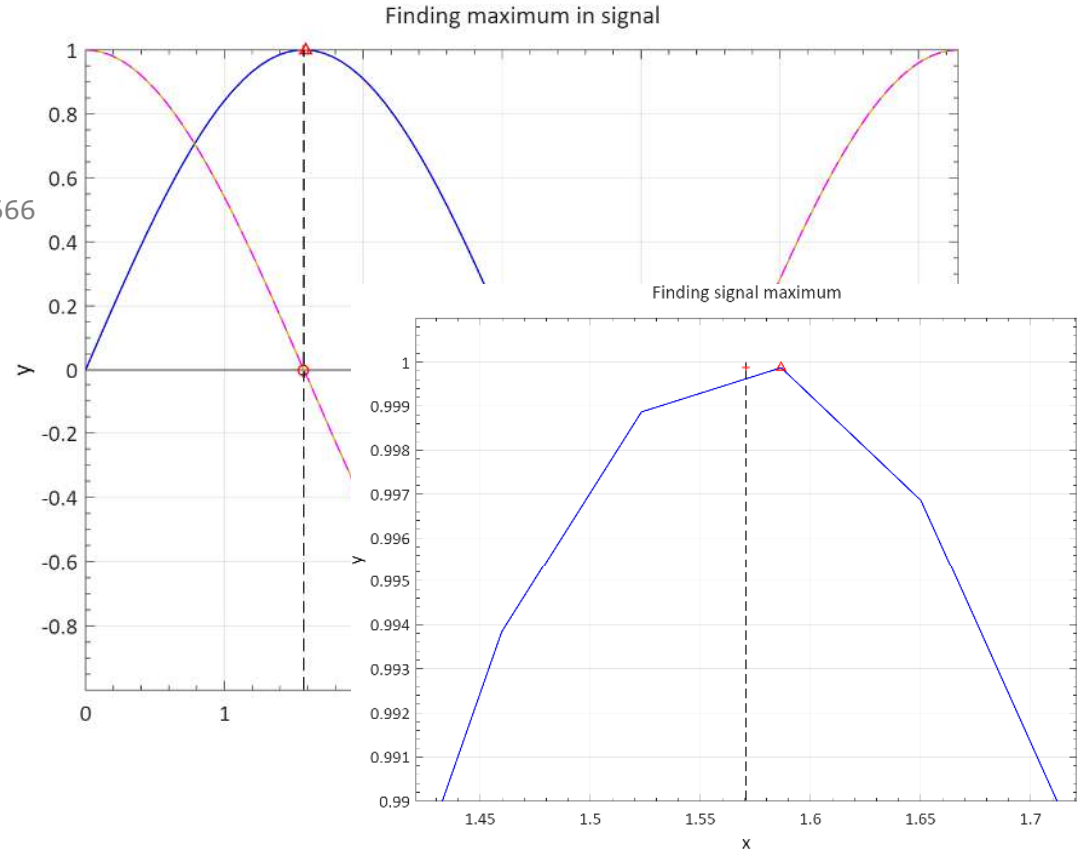
ylim([0.99, 1.001]);

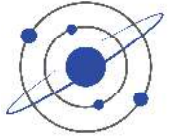
end

X_label('x');

Y_label('y');

Title('Finding maximum in signal');





Scientific-algorithmic programming

% Use of parabolic fit

```
kk = k1 + [-2:2];
```

```
plot(x(kk), y(kk), 'm*');
```

```
[coeffs, xmax, ymax] = ParabolicFit(x(kk), y(kk));
```

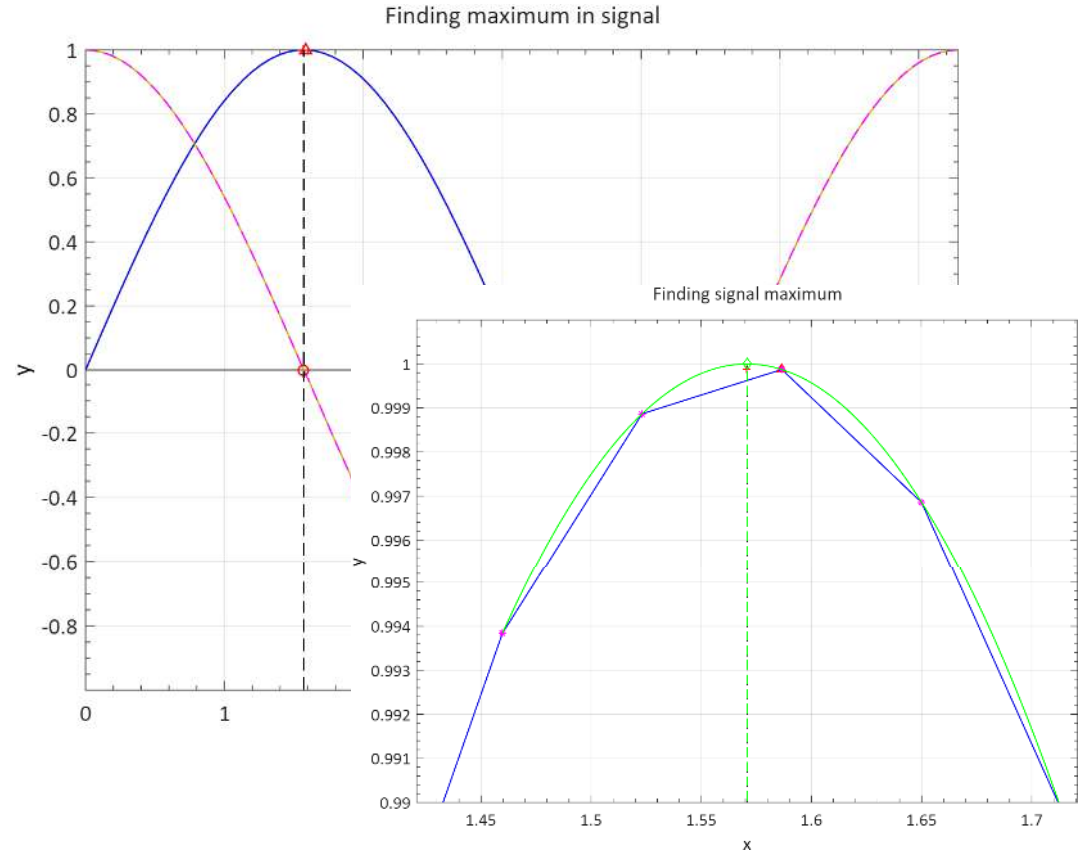
```
[ xmax , pi/2 ] % 1.570808 1.570796
```

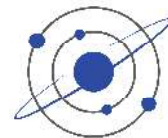
```
[ ymax , 1 ] % 0.999998 0.9999982
```

```
xx = linspace(x(kk(1)), x(kk(end)));
```

```
plot(xx, polyval(coeffs, xx), 'g-');
```

```
plot([xmax, xmax], [0, ymax], 'gd--')
```





Scientific-algorithmic programming

```
%--- Use of cubic fit
%  $y = a x^3 + b x^2 + c x + d$ 
coeffs3 = CubicFit(x(kk), y(kk));
a = coeffs3(1);
b = coeffs3(2);
c = coeffs3(3);
d = coeffs3(4);
yy3 = a * xx.^3 + b * xx.^2 + c * xx + d;
% == polyval(coeffs3, xx)
plot(xx, yy3, 'c-');

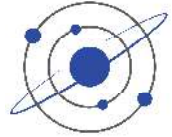
% Maximum @  $dy/dx == 0$ 
%  $dy/dx = 3 a x^2 + 2 b x + c == 0$ 
A = 3 * a;
B = 2 * b;
C = c;
% → solution using Quadratic Equation:
%  $x_0 = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$ 
x0 = (-B + sqrt(B^2 - 4 * A * C)) / (2 * A) % ≈ 127.58 (NO)
x0 = (-B - sqrt(B^2 - 4 * A * C)) / (2 * A) % ≈ 1.57077
x_max3 = x0;
y_max3 = polyval(coeffs3, x_max3)
plot([x_max3, x_max3], [0, y_max3], 'cd--')
```

```
%--- Use of interpolation
yy = interp1(x(kk), y(kk), xx, 'cubic');
%yy = interp1(x(kk), y(kk), xx, 'spline');
%yy = interp1(x(kk), y(kk), xx, 'makima');
plot(xx, yy, 'm-');
[y_max3, k3] = max(yy);
% here maximum depends on sampling...
x_max3 = xx(k3)
plot(x_max3 * [1, 1], [0, y_max3], 'm+--')
```

MATLAB



Matlab Online access



- How to link
 - https://login.mathworks.com/embedded-login/signin.html?cid=mldo&locale=en_US&uri=https%3A%2F%2Fdrive.mathworks.com%2Ffiles
 - Sign In: rlachance@gmail.com
- <https://drive.mathworks.com/files/>

→ Only 20 hours free...

drive.mathworks.com/files/

MATLAB Drive

Open editor

Files

Files

Shared Content

Deleted Files

Open in MATLAB

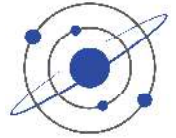
Upload

New Folder

Download

Name	Size
Examples	
General_Tools	
resources	
SPACE	
TIRS_Tools	
TIRS.prj	1 KB
TIRS-Rotations.m	24 KB

Working environment



matlab.mathworks.com

Apps SOLIDS Gmail Gmail2 Teams The Urantia Book Le Livre d'Urantia Bing Maps LinkedIn Météo Radar Z-Library All Bookmarks

HOME PLOTS APPS FIGURE FORMAT TOOLS Search (Ctrl+Shift+Space) Richard L

New Script New Live Script New Open Find Files Import Data Save Workspace Clean Data New Variable Open Variable Clear Workspace Favorites Clear Commands Simulink Layout Set Path Add-Ons Settings Help Feedback Learn MATLAB Request Support

FILE / > MATLAB Drive >

Fila

Name

Examples

General_Tools

resources

SPACE

TIDS Tools

Workspace

Name	Value	Size
alpha	-48.3118925...	1x1
alpha_X	-90	1x1
alpha_Y	45	1x1
alpha_Z	10	1x1
ans	2	1x1
arc	3x100 double	3x10
az	-90	1x1
A	1x101 double	1x10
beta	48.3118925...	1x1
B	1x101 double	1x10
city1	'Miami, USA'	1x10
city2	'Greenwich'	1x9
coastlat	9865x1 dou...	9865

Perturbed_vs_Unperturbed_Orbit_Trajectory_Propagation.mlx TIRS-Rotations.m Drawl Figure 1 x

/MATLAB Drive/TIRS-Rotations.m

```
91 %% 1.3) Great Circle path
92 % Shortest path from two points along the globe
93 % -> see STUDY_Loxodrome.m for many equivalent methods of computing the
94
95 city1 = 'Miami, USA';
96 lat1 = DMS_to_deg([25,45,48.85]); % °, ', ''
97 lon1 = -DMS_to_deg([80,07,46.72]);
98
99
100 % city2 = 'St-Malo, France';
101 % lat2 = DMS_to_deg([48,38,31.93]);
102 % lon2 = -DMS_to_deg([2,01,54.07]);
103 observer = DefineObserverLocation('Greenwich');
104 city2 = observer.Name;
105 lat2 = observer.Lat;
106 lon2 = observer.Lon; % =0 @ Greenwich
107
108 % 1) usig GreatCircle2 based on two coordinates on the globe ( $\lambda_1, \phi_1$ ) ->
109 GC = GreatCircle2(lat1, lon1, lat2, lon2); % between points
GC.Pos = GC.xyz * Earth.r1_km;
```

Command Window

>> |

Octave

- “MATLAB’s twin”
→ GNU open-source $\equiv$ FREE



GNU Octave “MATLAB’s twin”



- Is GNU Octave Any Good? A Free MATLAB Alternative Review
 - www.youtube.com/watch?v=JhR853Q5G94&t=28s
- MATLAB vs OCTAVE: A Comprehensive Comparison [2:31]
 - www.youtube.com/watch?v=M046hU_iME8&list=PLciHLArTqxSqBpEm9wdVgpUni-prgbk_-
- Packages
 - <https://octave.sourceforge.io/packages.php>

Category	Score	
Cost	★★★★★	← FREE is the biggest advantage
MATLAB compatibility	★★★★☆	← Easy to handle when we know the small syntax difference and missing functions to avoid or replace
Packages/Toolboxes	★★★☆☆	
IDE	★★★★☆	← Quite similar to MATLAB
Debugging	★★★★★	
Figures/Graphics	★★★☆☆	← <i>Maybe greatest disappointment...</i>
Execution speed	★★★★☆	← Not that critical for development, study & research

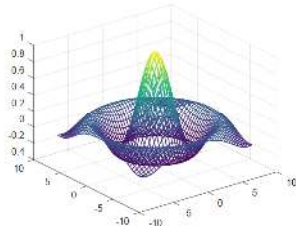
GNU Octave Installation



<https://octave.org>



[About](#) [Donate](#) [Download](#) [Get Involved](#) [Support](#)



GNU Octave
Scientific Programming Language

- Powerful mathematics-oriented syntax with built-in 2D/3D plotting and visualization tools
- Free software, runs on GNU/Linux, macOS, BSD, and Microsoft Windows
- Drop-in compatible with many Matlab scripts

</> Syntax Examples

The Octave syntax is largely compatible with Matlab. The Octave interpreter can be run in [GUI mode](#), as a console. Octave examples can be found in [the Octave wiki](#).

Solve systems of equations with linear algebra operations on **vectors** and **matrices**.

```
b = [4; 9; 2] # Column vector
A = [ 3 4 5;
      1 3 1;
      3 5 9 ]
x = A \ b # Solve the system Ax = b
```

Download

GNU Octave 10.3.0 is the latest stable release. ([Release Notes](#))

[Source](#) [GNU/Linux](#) [BSD](#) [macOS](#) [MS Windows](#)

</> Source

The latest released version of Octave is always available from

- <https://ftp.gnu.org/gnu/octave>
- <ftp://ftp.gnu.org/gnu/octave>

Please download from <https://ftpmirror.gnu.org/octave>, which will redirect automatically to a nearby mirror

Microsoft Windows

Note: All installers below bundle several Octave packages so [Read more](#).

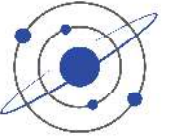
- Windows-64 (recommended)
 - [octave-10.3.0-w64-installer.exe](#) (~ 510 MB) [signature]
 - [octave-10.3.0-w64.7z](#) (~ 425 MB) [signature]
 - [octave-10.3.0-w64.zip](#) (~ 760 MB) [signature]

→ See instructions in file: [PHY-2100 Sciences de l'espace, Installation Octave.pdf](#)

Python

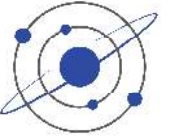


Python libraries/packages



- Python has many **useful/primordial** support packages
 - NumPy, for basic numerical routines <https://numpy.org/>
 - Astropy, for physical units and time handling
 - Poliastro → <https://docs.poliastro.space/en/stable/quickstart.html>
 - matplotlib, for orbit plotting <https://matplotlib.org/>
 - PyVista, for 3D plotting <https://pyvista.org/>
 - plotly, for 2D and 3D interactive orbit plotting
 - SciPy, for root finding and numerical propagation
 - Skyfield — used for getting star data and calculating star positions
 - jplephem, for the planetary ephemerides using SPICE kernels
 - numba, for accelerating the code

Python for astrodynamics



Sky Map in Python

→ <https://viyaleta.medium.com/how-to-make-a-sky-map-in-python-a362bf722bb2>

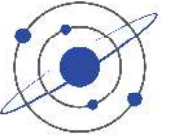
■ Prerequisites

- First, you need a version of Python installed on your machine. I am working out of Python 3.8.8
Then, we need the following libraries. I have listed them with the specific version I used in the demo.
- Numpy 1.19.2 — used for working with arrays of star data
- Matplotlib 3.3.4 — used for making the final star map image
- **Skyfield** 1.45 — used for getting star data and calculating star positions
- Datetime — used for converting string dates into date-time objects

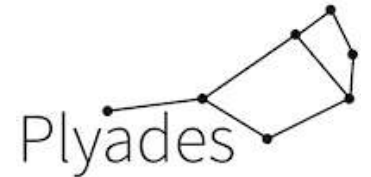
The following libraries are optional but useful if like me, you are also getting old, lazy, and cannot be bothered:

- Geopy 2.2.0 — converts our location into latitude and longitude
- Tzwhere — grabs the timezone of our latitude and longitude
- Pytz 2021.3 — used for converting date-time from local timezone into UTC timezone (which is needed for Skyfield to build the sky map properly)

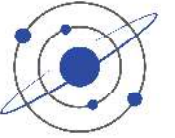
Astrodynamics libraries in Python



- **Astropy:** *A Community Python Library for Astronomy* www.astropy.org
 - Astropy is a core Python library designed for astronomy and astrophysics, fostering an ecosystem of interoperable astronomy packages
 - It provides a unified, open-source set of tools for handling physical units, coordinates, celestial time, cosmological calculations, and reading common astronomical file formats
- **Poliastro** www.poliastro.space; <https://docs.poliastro.space/en/stable/quickstart.html>
 - An open source (MIT) pure Python library for interactive astrodynamics and orbital mechanics, with a focus on ease of use, speed, and quick visualization. It provides a simple and intuitive API, and handles physical quantities with units.
 - Some features include orbit propagation, solution of the Lambert's problem, conversion between position and velocity vectors and classical orbital elements and orbit plotting, among others. It focuses on interplanetary applications, but can also be used to analyze artificial satellites in Low-Earth Orbit (LEO)
- **Plyades:** *A Python Astrodynamics Library* <https://plyades.readthedocs.io>
 - An MIT-licensed astrodynamics library written in Python. It aims to provide a comprehensive toolset for fast development of performant mission analysis applications
 - The API provides powerful high-level objects for pythonic ease-of-use while the low-level functional building blocks can also be used independently



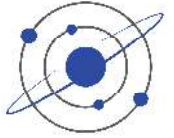
Astrodynamics libraries in Python



- **Skyfield** *Elegant Astronomy for Python* <https://rhodesmill.org/skyfield>
 - An elegant, pure-Python astronomy library used to compute high-precision, research-grade positions for stars, planets, and Earth-orbiting satellites
 - Its calculations are exceptionally accurate, designed to match the predictions of the United States Naval Observatory and the Astronomical Almanac to within 0.0005" (half a "mas" or milliarcsecond)
 - Written in pure Python and installs without any compilation
 - Supports Python 2.6, Python 2.7 and Python 3
 - Has NumPy as its only binary dependency, the fundamental package for scientific computing with Python, whose vector operations make Skyfield efficient
- **ESA Godot** <https://godot.io.esa.int/docs/guides/index.html>
 - GODOT is a set of libraries which perform many tasks related to astrodynamics.

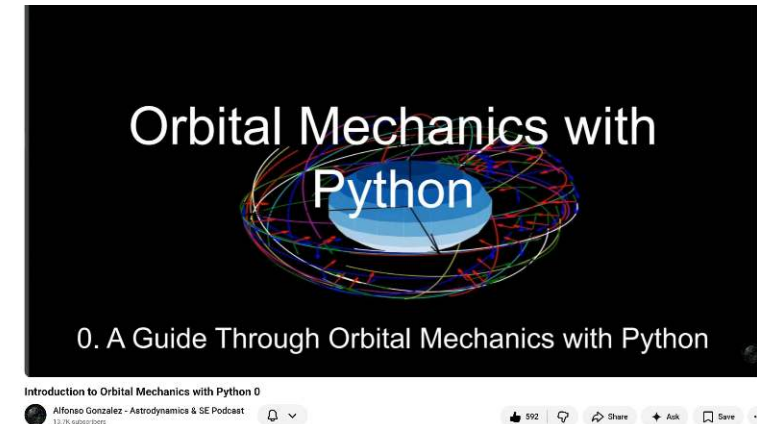


Astrodynamics libraries in Python



- **Alfonso Gonzalez AWP - Astrodynamics with Python**
 - <https://github.com/alfonsogonzalez/AWP>
 - www.youtube.com/@alfonsogonzalez-astrodynam2207

- **Introduction to Orbital Mechanics with Python 0**
 - Alfonso Gonzalez – Astrodynamics
www.youtube.com/watch?v=mEbLsoryfHo&t=100s [06:27]

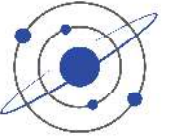


References

- Key books
- Search tools
- Web resources



References



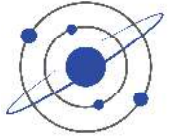
This reference material list covers orbital mechanics and astrodynamics, with a focus on Mechanical or Aerospace engineering undergraduate students. This list, in alphabetical order, is far from being exhaustive.

- **Baker** 1967, [Astrodynamics Applications and Advanced Topics](#)
- **Bate** 1971, [Fundamentals of Astrodynamics](#) + 2nd Ed. 2020
- **Battin** 1999, [An Introduction to the Mathematics and Methods of Astrodynamics](#) (Revised Ed.)
- **Boulet** 1991, [Methods of orbit determination for the microcomputer](#)
- **Campbell** 1996, [Introduction to Space Sciences and Spacecraft Applications](#)
- **Celletti** 2007, [The Waltz of the Planets](#)
- **Chobotov** 2002, [Orbital Mechanics](#) (3rd Ed.)
- **Colwell** 1993, [Solving Kepler's Equations Over three Centuries](#)
- **Curtis**, 2020, [Orbital Mechanics for Engineering Students](#) (4th Ed.)
Teaches students the basics of orbital mechanics and their application in engineering.
Includes a range of exercises and examples to help readers understand the concepts covered.
- **Danby** 1992, [Fundamentals of Celestial Mechanics](#) (2nd Ed.)
- **Eagle** 2012, [Celestial Computing with MATLAB](#)
- **Fitzpatrick** 2012, [An Introduction to Celestial Mechanics](#)



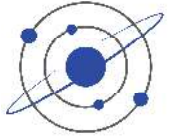
Do not hesitate to
buy used versions
of some of these
books on Amazon
for only a few \$!

References



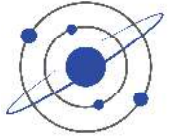
- **Feuerbacher** 2006, *Utilization of Space; Today and Tomorrow*
- **Fortescue** 2011, *Spacecraft Systems Engineering* (4th Ed.)
- **Gleisner** 2013, “Three solutions to the 2-Body Problem”, Degree project, Linnaeus University, Sweden
- **Heafner** 1999, “Fundamental Ephemeris Computations: For use with JPL data”
- **Herrick** 1971, *Astroynamics*, Vol. 1 – Orbit Determination, Space Navigation, Celestial Mechanics
Vol. 2 – Orbit correction, Perturbation theory, Integration
- **Hintz**, 2015, *Orbital Mechanics and Astroynamics; Techniques and Tools for Space Missions*
→ Table 2.1 p.53 for list of useful relations
- **Hughes** 2004, *Spacecraft Attitude Dynamics*
- **Humble** 1995, *Space Propulsion Analysis and Design*
- **Kluever** 2018, *Space Flight Dynamics*
- **Larson** 1999, *SMAD III – Space Mission Analysis and Design* (3rd Ed.)
- **Lawrence** 2018, *Celestial Calculations; A Gentle Introduction to Computational Astronomy*
- **Lissauer** 2023, *Fundamental Planetary Science Physics, Chemistry and Habitability*
- **Liu** 2023, *Algorithms for Satellite Orbital Dynamics*
- **Meeus** 1997, *Mathematical Astronomy Morsels* + 2002, *More Mathematical Astronomy Morsels*

References



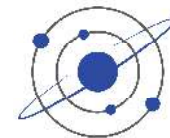
- **Meeus** 1998, *Astronomical Algorithms* (2nd Ed.)
provides detailed algorithms for calculating various celestial phenomena, such as planetary positions, eclipses, and lunar phases
→ This is an excellent reference source for many relevant calculations in orbital mechanics and is highly recommended
- **Montenbruck** 2000, *Astronomy on the Personal Computer* (4th Ed.)
- **Montenbruck** 2005, *Satellite Orbits: Models, Methods and Applications*
Comprehensive guide to satellite orbits, covering topics such as orbit determination, perturbations, and satellite constellations.
- **Moulton** 1914, *An Introduction to Celestial Mechanics* (2nd Ed Rev.)
- **Murray** 1999, *Solar System Dynamics* <https://assets.cambridge.org/97805215/72958/sample/9780521572958wsn01.pdf>
- **Pisacane** 2005, *Fundamentals of Space Systems* (2nd Ed) → Chapter 2: The Space Environment
- **Prussing** 2013, *Orbital Mechanics* (2nd Ed.)
- **Purohit** 2016, *A Guide to Astronomical Calculations with Illustrative Solved Examples*
- **Schaub** 2009, *Analytical Mechanics of Space Systems* (2nd Ed.)
- **Sellers** 2015, *Understanding Space: An Introduction to Astronautics* (4th Ed.)
Provides a clear, foundational overview of spaceflight, covering key concepts in orbital mechanics, rocket propulsion, and spacecraft design. It is designed as an introductory textbook for students and professionals interested in space science and astronautics.
- **Sidi** 2000, *Spacecraft Dynamics and Control: A Practical Engineering Approach*
- **Schmidt** 2021, *Numerical Python in Astronomy and Astrophysics; A Practical Guide to Astrophysical Problem Solving*
- **Spencer** 2023, *Interplanetary Astrodynamics*

References



- **Sutton** 2016, [Rocket Propulsion Elements \(9th Ed.\)](#)
Comprehensive guide to rocket propulsion, covering topics such as rocket engines, propellants, and performance analysis.
- **Taff** 1981, [Computational Spherical Astronomy](#)
- **Taff** 1985, [Celestial Mechanics: A Computational Guide for the Practitioner](#)
- **Tewari** 2007, [Atmospheric and Space Flight Dynamics: Modeling and Simulation with MATLAB and Simulink](#)
- **Tewari** 2021, [Foundations of Space Dynamics](#)
- **Thomson** 1986, [Introduction to Space Dynamics](#)
- **Urban** 2012, [Explanatory Supplement to the Astronomical Almanac \(3rd Ed.\)](#)
- **Vallado** 2013, [Fundamentals of Astrodynamics and Applications \(4th Ed.\)](#)
Comprehensive guide to astrodynamics, covering everything from orbital mechanics to spacecraft formation flying. It is a popular textbook for students and professionals alike, offering clear explanations and numerous examples.
- **Walter** 2018, [Astronautics: The Physics of Space Flight](#)
- **Wertz** 1999, [Space Mission Analysis and Design \(3rd Ed.\)](#)
Covers the design of space missions, including mission analysis, system design, and risk analysis. It is a comprehensive guide for professionals in the aerospace industry.
- **Wertz** 2001, [Orbit and Constellation Design and Management](#)
Comprehensive guide on the technical and strategic aspects of satellite orbits and the design of constellations. Covers principles of orbital mechanics, mission planning, and the optimization of satellite constellations for global coverage and communication systems.

J.J. Sellers courses (FAA)



FAA Chapters

 III.4.1.1_Space_in_our_lives.pdf

 III.4.1.2_The_Space_Environment.pdf

 III.4.1.3_Understanding_Orbits.pdf

 III.4.1.4_Describing_Orbits.pdf

 III.4.1.5_Maneuvering_in_Space_Hohmann2.pdf

 III.4.1.6_Interplanetary_Travel.pdf

 III.4.1.7_Returning_from_Space.pdf

 III.4.3.1_Space_Vehicle_Control_Systems.pdf

 III.4.4.1_Environmental_Control_and_life_support.pdf

 Sellers 2004, Understanding Space; An Introduction to Astronautics (2nd Ed).pdf - Shortcut

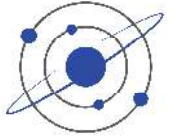
 Sellers 2015, Understanding Space; An Introduction to Astronautics (Revised 2nd Ed).pdf - Sho...

www.faa.gov/sites/faa.gov/files/about/office_org/headquarters_offices/avs/III.4.2.1_Rockets_and_Launch_Vehicles.pdf

Web reference for getting scientific papers (and books)



Useful web sources for books and scientific papers



1) Use **Google Scholar** (<https://scholar.google.ca>) to see if the paper is not already freely available

e.g.: “Murphy 1967, HARP 5-Inch and 16-Inch Guns at Yuma Proving Ground, Arizona”

→ The Best Google Scholar Alternatives that Even Your Supervisor Doesn’t Know About: www.youtube.com/watch?v=jnJv9nl95UQ

2) Use Laval University resources... =?

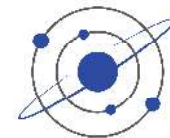
3) Use **advanced search tools** for books, articles & scientific research papers

→ Get ANY Research Paper in 10 s: https://youtu.be/C-hHQ5shJLk?si=tGCalcExX_BGGYl1

→ Sci-Hub Alternatives | The Research Shortcut: 6 Hidden Platforms Every Student Needs (2025):
www.youtube.com/watch?v=1OkAw19pqK4&t=243s

Web references





https://en.wikipedia.org/wiki/Orbital_mechanics

Part of a series on
Astrodynamics



Orbital mechanics

Orbital elements [hide]

Apsis · Argument of periaxis · Eccentricity ·
Inclination · Mean anomaly · Orbital nodes ·
Semi-major axis · True anomaly

Types of **two-body orbits** by [hide] eccentricity

Circular orbit · Elliptic orbit
Transfer orbit
(Hohmann transfer orbit · Bi-elliptic transfer orbit)
Parabolic orbit · Hyperbolic orbit · Radial orbit ·
Decaying orbit

Equations [hide]

Dynamical friction · Escape velocity ·
Kepler's equation ·
Kepler's laws of planetary motion ·
Orbital period · Orbital velocity ·
Surface gravity · Specific orbital energy ·
Vis-viva equation

Celestial mechanics

Gravitational influences [hide]

Barycenter · Hill sphere · Perturbations ·
Sphere of influence

N-body orbits [hide]

Lagrangian points
(Halo orbits)
Lissajous orbits · Lyapunov orbits

Engineering and efficiency

Preflight engineering [hide]

Mass ratio · Payload fraction ·
Propellant mass fraction ·
Tsiolkovsky rocket equation

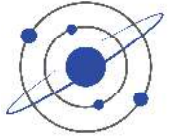
Efficiency measures [hide]

Gravity assist · Oberth effect

Propulsive maneuvers [hide]

Orbital maneuver · Orbit insertion

Introduction to Orbital Mechanics – Lynnane George



<https://oer.pressbooks.pub/lynnanegeorge/>

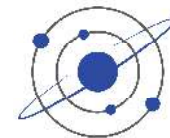
<https://colorado.pressbooks.pub/introorbitalmechanics/front-matter/introduction/>

Introduction

1. [Chapter 1 - Orbital Basics](#)
2. [Chapter 2 - Orbit Geometry](#)
3. [Chapter 3 - The Classical Orbital Elements \(COEs\)](#)
4. [Chapter 4 - Preliminary Orbit Determination](#)
5. [Chapter 5 - Launch Windows and Time](#)
6. [Chapter 6 – Kepler’s Prediction Problem](#)
7. [Chapter 7 - Maneuvering](#)
8. [Chapter 8 - Rendezvous](#)
9. [Chapter 9 - Ground Tracks](#)
10. [Chapter 10 - Orbital Perturbations](#)
11. [Chapter 11 - Interplanetary Travel](#)



Astronomy



- Pressbooks, Campus, Douglas College, New Westminster, British Columbia, Canada → Jennifer Kirkey

- <https://pressbooks.bccampus.ca/astronomy1105/chapter/1-0-introduction/>
- This book can be downloaded for free at <https://openstax.org/details/books/astronomy>



BRITISH COLUMBIA/YUKON
PRESSBOOKS

- State University of New York (SUNY) Open Educational Resources (OER) Services

- <https://courses.lumenlearning.com/suny-astronomy/>

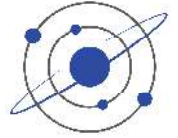


OER
services

Astronomy

- Chapter 1: Science and the Universe: A Brief Tour
- Chapter 2: Observing the Sky: The Birth of Astronomy
- Chapter 3: Orbits and Gravity
- Chapter 4: Earth, Moon, and Sky
- Chapter 5: Radiation and Spectra
- Chapter 6: Astronomical Instruments
- Chapter 7: Other Worlds: An Introduction to the Solar System
- Chapter 8: Earth as a Planet
- Chapter 9: Cratered Worlds
- Chapter 10: Earthlike Planets: Venus and Mars
- Chapter 11: The Giant Planets
- Chapter 12: Rings, Moons, and Pluto
- Chapter 13: Comets and Asteroids: Debris of the Solar System
- Chapter 14: Cosmic Samples and the Origin of the Solar System
- Chapter 15: The Sun: A Garden-Variety Star
- Chapter 16: The Sun: A Nuclear Powerhouse
- Chapter 17: Analyzing Starlight
- Chapter 18: The Stars: A Celestial Census
- Chapter 19: Celestial Distances
- Chapter 20: Between the Stars: Gas and Dust in Space
- Chapter 21: The Birth of Stars and the Discovery of Planets outside the Solar System
- Chapter 22: Stars from Adolescence to Old Age
- Chapter 23: The Death of Stars
- Chapter 24: Black Holes and Curved Spacetime
- Chapter 25: The Milky Way Galaxy
- Chapter 26: Galaxies
- Chapter 27: Active Galaxies, Quasars, and Supermassive Black Holes
- Chapter 28: The Evolution and Distribution of Galaxies
- Chapter 29: The Big Bang
- Chapter 30: Life in the Universe
- Appendices

Introduction to Aerospace Flight Vehicles – Leishman 2025



- Leishman 2025, Introduction to Aerospace Flight Vehicles

- <https://eaglepubs.erau.edu/introductiontoaerospaceflightvehicles/>

- I. Introductory

- II. Preparations

- III. Design Aspects of Flight Vehicles

- IV. Fluid Properties & Hydrostatics

- V. Introduction to Fluid Dynamics

- VI. Modeling of Fluid Motion

- VII. Other Types of Flows

- VIII. Airfoil & Wing Aerodynamics

- IX. Introduction to Airfoil & Wing Theory

- X. Basis of Airplane Performance

- XI. Propulsion Systems

- XII. Atmospheric Flight Performance

- XIII. Faster, Higher, & Into Space** → XIII – 52. Supersonic Flight Vehicles

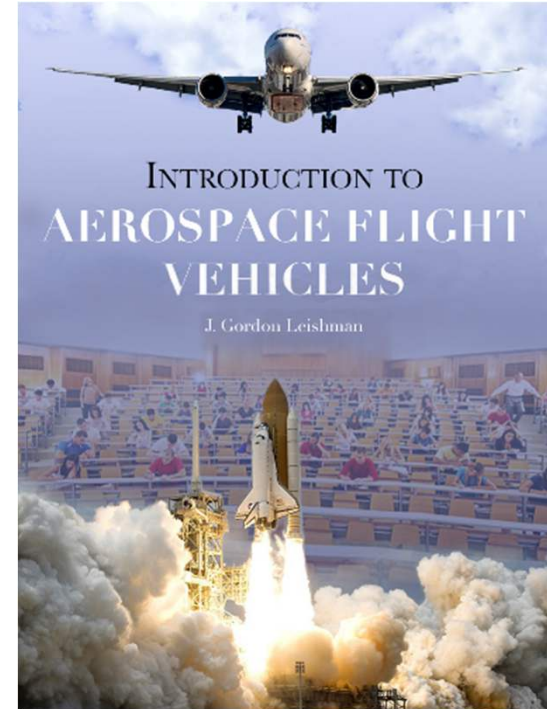
- XIV. Advanced Topics

- XV. Other Types of Flight Vehicles

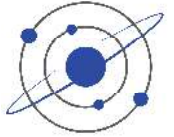
- XVI. Worked Problems & Examples

- Essential Formulae

- 52. Supersonic Flight Vehicles
 - 53. Hypersonic Flight Vehicles
 - 54. Rockets & Launch Vehicles
 - 55. Astronautics, Space & Astrodynamics



U.S. Naval Observatory – Astronomical Information Center



<https://aa.usno.navy.mil/faq/index>

Terms used on this website

- [Astronomical Almanac Glossary](#)

Phenomena of the Sun and Moon

- [Rise, Set, and Twilight Definitions](#)
- [Comparative Lengths of Days and Nights](#)
- [Length of Day and Night at the Equinoxes](#)
- [Sunrise and Sunset Times Near the Solstices](#)
- [Visibility of the Crescent Moon](#)
- [References on Eclipses](#)
- [Phases of the Moon and Percent of the Moon Illuminated](#)

Time

- [Universal Time](#)
- [Terrestrial Time](#)
- [Approximate Sidereal Time](#)
- [U.S. Time Zones](#)
- [World Time Zone Map](#)
- [Daylight Saving Time](#)
- [The Equation of Time](#)

Calendars and Historical Events

- [Introduction to Calendars](#)
- [Leap Years](#)
- [The International Date Line](#)
- [The 21st Century and the 3rd Millennium](#)
- [The First Sunrise of the New Year](#)

Computing Astronomical Data and Phenomena

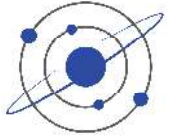
- [Converting Between Julian Dates and Gregorian Calendar Dates](#)
- [Computing Times of Rise, Set, and Twilight](#)
- [Computing Approximate Solar Coordinates](#)
- [Computing Approximate Sidereal Time](#) → GMST
- [Computing Altitude and Azimuth](#)

Celestial Navigation

- [USNO Celestial Navigation Algorithms](#)
- [Celestial Navigation Resources](#)

Other Topics

- [Astronomical Data Used for Litigation](#)
- [The Seasons and the Earth's Orbit](#)
- [Variation in Times of Perihelion and Aphelion](#)
- [International Celestial Reference System \(ICRS\)](#)
- [When Did the Asteroids Become Minor Planets?](#)



<https://celestrak.org/columns/>

Computers & Satellites, By Dr. T.S. Kelso

[Orbital Propagation, Part I](#)

[Orbital Propagation, Part II](#)

[Orbital Estimation](#)

[Orbit Determination](#)

[Orbital Coordinate Systems, Part I](#)

[Orbital Coordinate Systems, Part II](#)

[Orbital Coordinate Systems, Part III](#)

[System Benchmarking](#)

[Visually Observing Earth Satellites. Real-World Benchmarking. Tracking the Sun and Moon](#)

[Sunrise . . . Sunset](#)

[Software Review: Micro Orbiter 3.0](#)

[Space Surveillance](#)

[Frequently Asked Questions: Two-Line Element Set Format](#)

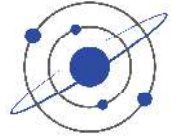
[More Frequently Asked Questions](#)

[Basics of the Geostationary Orbit](#)

[More on Geostationary Orbits](#)



Kerbal Space Program



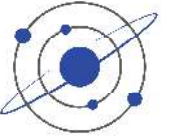
- TO BE COMPLETED



Acronyms



Acronyms



- → See updated list in file: [SpaceOrbitLAB, Astrodynamics Acronyms 2026-09-01.pdf](#)